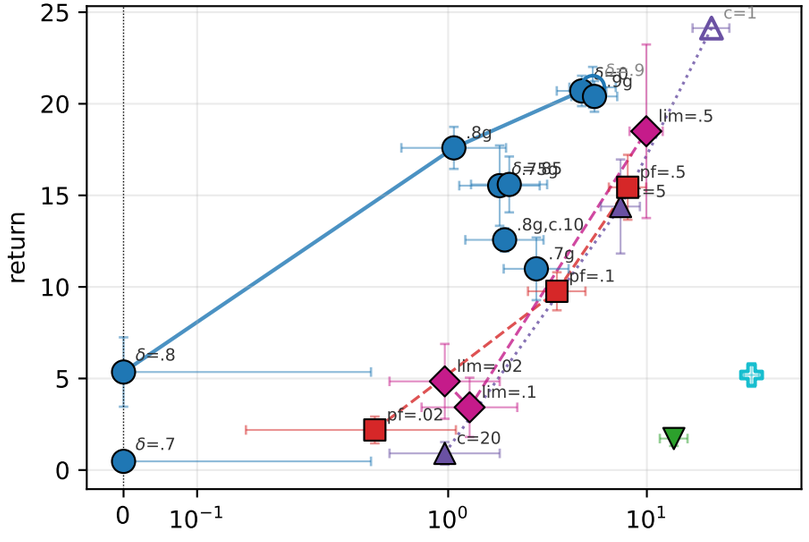
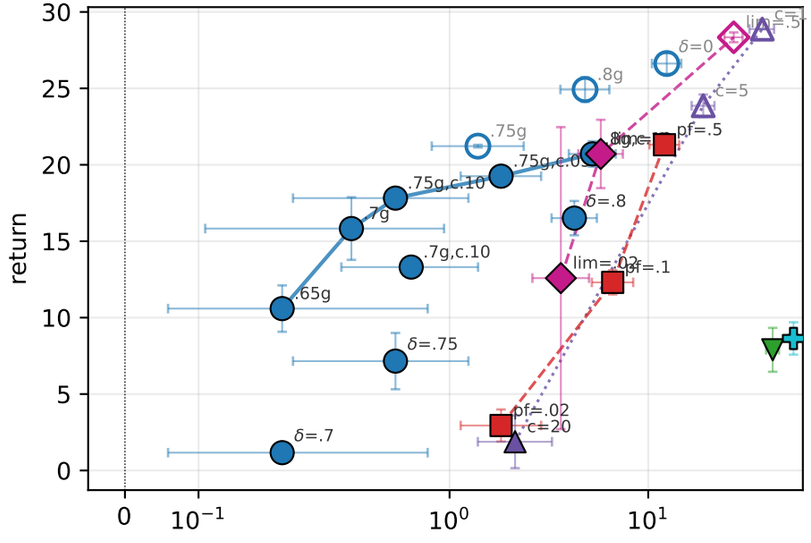
Return vs. violation rate (final checkpoint, 312 episodes/seed; hollow = soft cost > 25/episode)

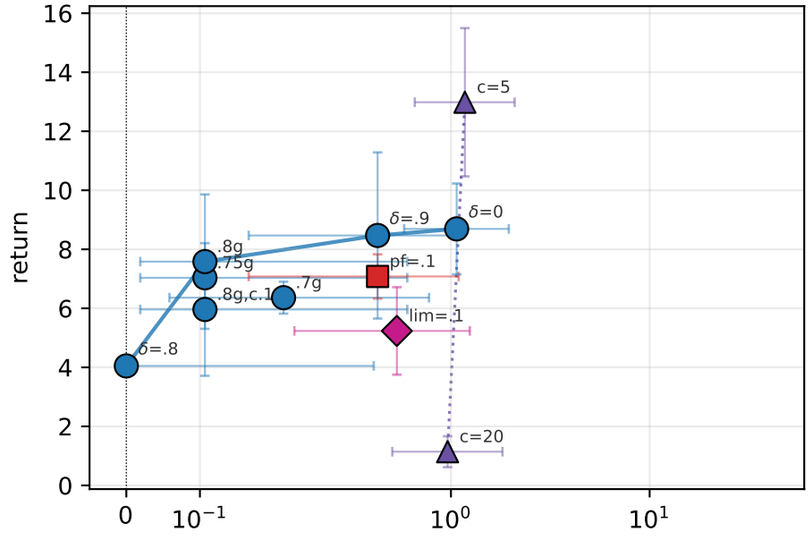
## PointGoal2



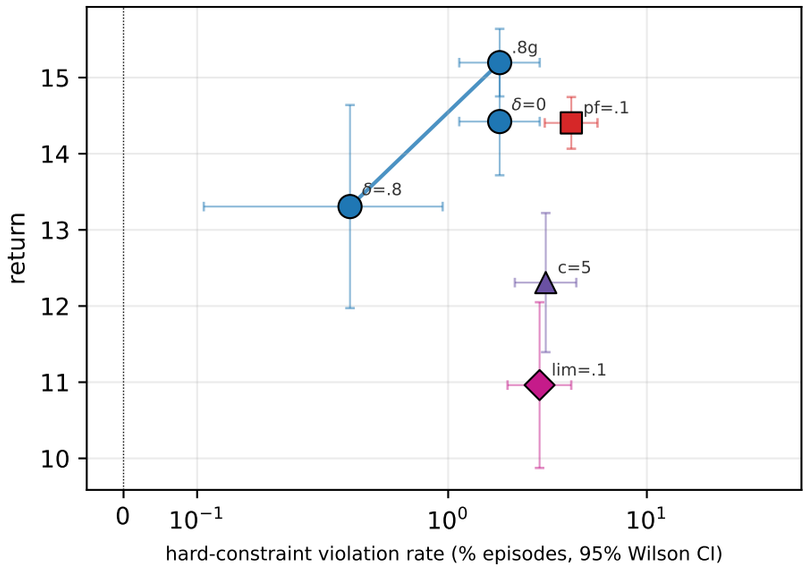

## CarGoal2



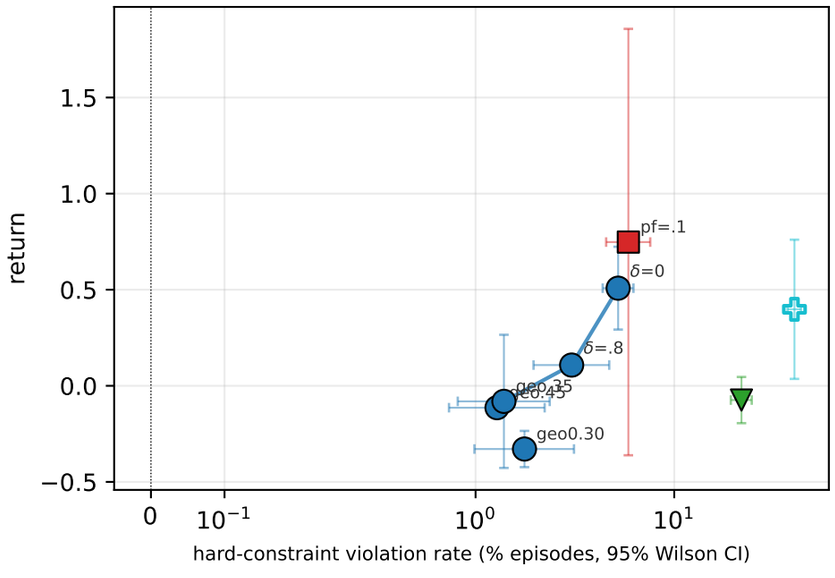

## PointPush2



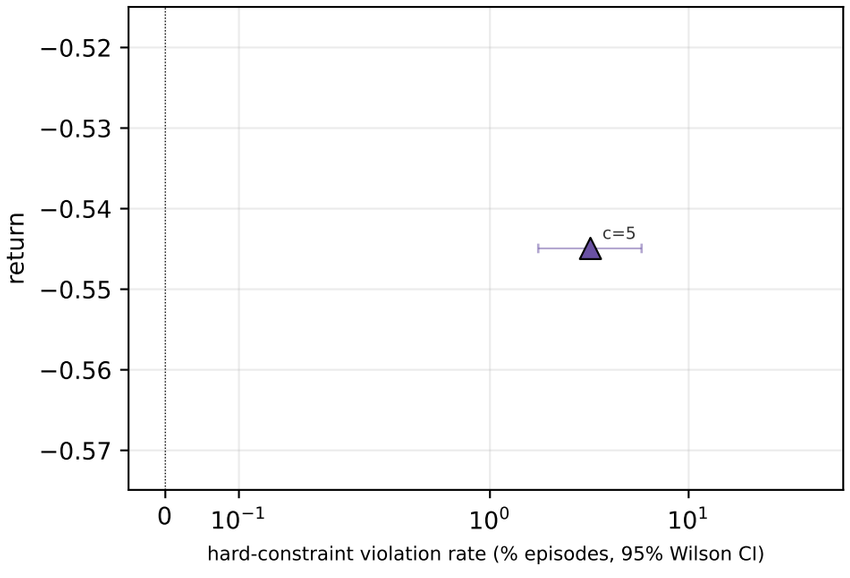

## CarPush2




## PointButton2




## CarButton2




SafeOT-typed (ours)
FDPI (constraint-based)
SAC-Lagrangian (penalty-based)
SAC + fixed penalty (penalty-based)
SDAC (multi-constraint SOTA)
SRCPO (multi-constraint SOTA)
typed Pareto frontier (soft ≤ 25)
SAC-Pen coefficient sweep
FDPI $p_f$ sweep
SAC-Lag cost-limit sweep